

AKASH CHINTHAMANIPETA

Robotics & Automation Platforms Engineer | Reusable Automation Architectures | Machine Vision & Embedded Controls
Denver, CO • akashchintamani24@gmail.com • +1 (303) 520-6428 • akashtechnologies.tech • [LinkedIn](https://www.linkedin.com/in/akashchintamani24/)

PROFESSIONAL SUMMARY

Robotics and automation engineer who takes multidisciplinary production systems from concept through prototyping, integration, factory validation, and deployment. M.S. in Mechanical Engineering (Robotics & Control), CU Boulder. Builds automation as reusable platforms rather than one-off machines: a common ESP32/Modbus control architecture, standardized I/O and safety patterns, and shared documentation now underpin filling, labeling, inspection, pasteurization, and water-management equipment across a production site. Hands-on across collaborative robotics, machine vision and 3D perception, autonomous navigation and sensor fusion, industrial controls, pneumatics, motion control, and OT-layer telemetry and alerting.

CORE COMPETENCIES

Platform & Architecture: Reusable control architectures, standardized I/O and comms patterns, machine design, commissioning, factory validation, deployment documentation, DFM

Robotics & Perception: Universal Robots cobots, Cognex AI vision, ROS2, motion planning (RRT), SLAM, LiDAR / IMU / optical-flow sensor fusion, stereo vision and structured-light depth estimation

Industrial Controls & Safety: State-machine logic, 24 VDC controls, emergency interlocks, alarm and fault-recovery logic, sensor-based inspection and rejection, pneumatics, stepper and BLDC motion control

Communications & IT/OT: Modbus RTU/RS485, MQTT, Ethernet I/O, I2C/SPI/UART, Home Assistant supervision, remote monitoring, webhook-based fault notification

Embedded & Software: ESP32 / ESP-IDF / ESPHome, LVGL touchscreen HMI, C/C++, Python, MATLAB/Simulink, LabVIEW, KiCad PCB design, SolidWorks, CATIA, AutoCAD, ANSYS

PROFESSIONAL EXPERIENCE

Robotics & Automation Engineer | Algae Barn — Colorado

Jun 2025 – Present

- Own end-to-end design, electrical integration, programming, commissioning, validation, and production support for custom automation equipment across an aquaculture manufacturing operation.
- Established a reusable embedded control platform — common ESP32/Modbus RS485 carrier boards, standardized RS485 pinouts and timing, shared I/O expansion, LVGL touchscreen HMI, and a common alerting pattern — now replicated across eight-plus production machines, cutting build time and support burden on each new deployment.
- Designed and deployed an automated four-station pod filling system with industrial flow sensing, pneumatic valves, embedded control, and production sequencing; eliminated a labor-intensive manual process with estimated savings of \$40K–\$50K.
- Delivered a second-generation phytoplankton filling and pressure-control platform (dual ESP32, Modbus RS485, capacitive level sensing, touchscreen HMI) that reduced fill time from 2.5 s to 1 s per jar and added an automated three-cycle clean-in-place routine with float-controlled fill/drain and completion notification.
- Self-initiated and built an automated cap labeling machine after observing hours of manual application: conveyors, stepper motion, machine vision with AI orientation and defect checking, pneumatic actuation, vacuum handling, and robot handshakes — sustaining roughly 450 caps/hour with automatic rejection of bad units.
- Built an automated jar rejection conveyor with sensor-based cap inspection, pneumatic transfer and reject mechanisms, full-line detection, alarm logic, and fault recovery.
- Engineered distributed water-management, RO, and pasteurization controls with 4–20 mA flow and pressure instrumentation, pump scheduling, level protection, emergency interlocks, scheduled purge cycles, and remote supervision.
- Instrumented equipment with condition telemetry and automated fault alerting to the operations team, converting reactive downtime into monitored, actionable events.
- Documented architectures, wiring, and operating procedures so machines can be run by operators without training and rebuilt or extended by others.

Research & Teaching Assistant | University of Colorado Boulder — Boulder, CO

Aug 2024 – May 2025

- Customized flight-controller firmware and integrated optical flow, LiDAR, MAVLink, and sensor fusion for GPS-denied autonomous navigation on a sub-75 g aerial robotics platform — directly transferable to indoor autonomous navigation and localization for mobile robots.
- Developed instructional materials and mentored students in computational methods, sensor integration, data acquisition, and engineering analysis (MCEN 3047).

Robotics Engineering Intern | Emerson — Boulder, CO

May 2024 – Aug 2024

- Designed and implemented an autonomous visual inspection cell for Coriolis mass flow sensor manufacturing using a Universal Robots collaborative robot, Cognex AI vision, and supervised defect classification.
- Implemented RRT-based motion planning and Ethernet I/O integration for automated brazing-paste defect inspection, improving inspection productivity by 30% with 95% defect-detection accuracy.
- Supported system testing, debugging, validation, and manufacturing integration with cross-functional engineering stakeholders.

SELECTED RESEARCH & TECHNICAL PROJECTS

- **Operator-Trainable Embedded Control Architecture** — Authoring a paper (targeting IEEE ICRA 2027) on a reconfigurable control architecture for small-scale production automation, covering fill accuracy, cycle time, operator usability, and reconfiguration without engineering support.
- **FactorySense** — ESP32-S3 industrial condition-monitoring platform for temperature, vibration, and current sensing with MQTT-based analytics and alerting; scoped and led a student engineering team through firmware and cloud workstreams.
- **HelioNode** — Low-cost solar-powered ESP32 environmental and soil-monitoring node; accepted for oral presentation at the 2026 IEEE Global Humanitarian Technology Conference.
- **Multi-Modal Depth Estimation** — Stereo vision, structured light, and IMU fusion with physics-based constraints; MAE 0.0117, RMSE 0.0241.

EDUCATION

M.S., Mechanical Engineering — Robotics & Control | University of Colorado Boulder 2023 – 2025
GPA 3.7/4.0

B.Tech., Mechanical Engineering | Jawaharlal Nehru Technological University, India 2019 – 2023
GPA 7.98/10 • Most Versatile Outgoing Student

PROFESSIONAL LEADERSHIP & SERVICE

- Secretary, IEEE Denver Section Computational Intelligence Society Chapter — technical programming, speaker engagement, and professional events.
- Technical Reviewer, IEEE Global Humanitarian Technology Conference 2026.
- Engineering Judge, FIRST Tech Challenge; Judge, CU Boulder ECEE First-Year Engineering Expo.
- Mentor to student robotics initiatives and engineering interns; active contributor to IEEE professional activities and outreach.